NETWORKING

Open Switch Platform Architecture (OSPA)

- A Unified Network Switch Platform Reference Design

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Problem Statement

Fragmented Software

While standards like SAI make data plane software portable, the switch platform layer remains proprietary and fragmented.

Slow Onboarding

Bringing up and onboarding new hardware platforms requires months of duplicated, custom engineering effort.

High Lifecycle Cost

A lack of standard hardware design multiplies long-term maintenance overhead and ongoing troubleshooting costs.

Why Now?



Shorter AI Cycles

AI infrastructure demands rapid deployment and a much shorter device onboarding cycle to keep pace with accelerating innovation.



Simultaneous Scaling

Modern network environments require onboarding many complex devices at the same time, multiplying execution complexity.



Outpaced Legacy Ways

Traditional, manual, and custom-engineered onboarding practices can no longer scale to meet these high-concurrency demands.

Open Switch Platform Architecture (OSPA)

Unified Architecture

Defines a unified architecture for the switch platform layer, aligning hardware and software interfaces.

Rapid Onboarding

Accelerates platform integration by promoting reusable hardware designs and leveraging established software stacks.

Optimized Costs

Minimizes long-term maintenance overhead and ongoing operational costs through standardized platform designs.

Scope of OSPA Specification

Included in Specification

Defines the switch platform layer design, encompassing critical management and interface functions:

- Power Management
- Sensor & environment control
- Inventory Management
- Out-of-band hardware interfaces
- Firmware maintenance
- Hardware platform troubleshooting

Excluded from Specification

Higher-level networking and control plane functions are outside the scope of this hardware-centric platform design:

- Routing protocols
- Forwarding management

OSPA vs. SAI vs. NOS

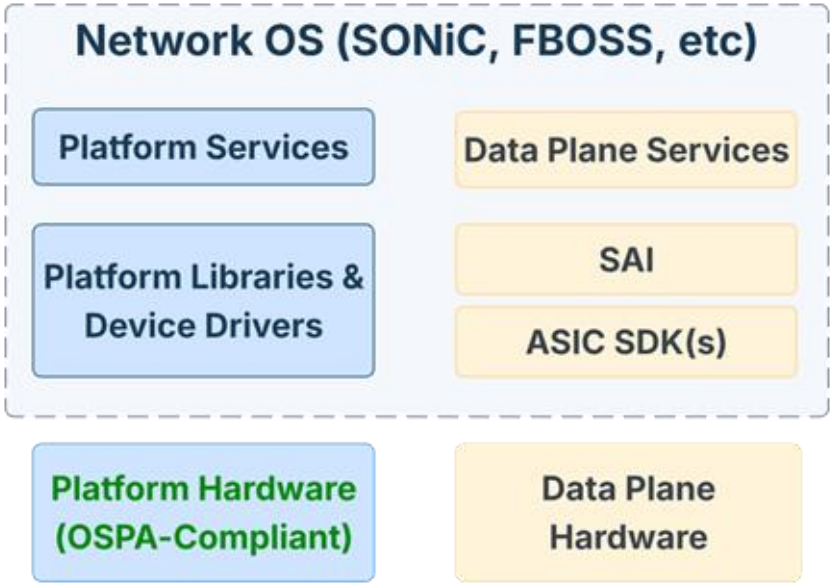
Key Principles

Platform Layer Specification

OSPA defines the platform layer architecture, independent from the data plane.

Open & Neutral

Completely vendor-neutral and NOS-neutral, fostering broad community adoption.



Industry Partners/Contributors

We would like to acknowledge the following companies for their valuable feedback and contributions (listed in alphabetical order):

- **Accton**
- **Arista**
- **Celestica**
- **Cisco**
- **DTI**
- **Micas**
- **Nexthop**
- **NVIDIA**
- **Wistron**

OSPA Specification - Latest Status

Specification Status

OSPA Specification 1.0

Submitted to OCP for official review.

Key Milestone Highlights:

- Complete baseline architectural definition
- Formal submission under OCP governance
- Ready for community feedback & review

Open Subproject Resources

The OSPA subproject is officially open for community collaboration:

Home Page
ocp-all.groups.io/g/ocp-ospa

Email Group
ocp-ospa@OCP-All.groups.io

GitHub Repository
https://github.com/opencomputeproject/OCP-switch-platform-architecture

Production Ready OSPA Switch Examples

FBOSS Support

Minipack3N

OSPA-Compliant open networking hardware platform with FBOSS support.

View Meta OCP Announcement

SONiC Support

Bailly CPO Switch

OSPA-Compliant co-packaged optics switch powered by SONiC.

View Micas Networks News

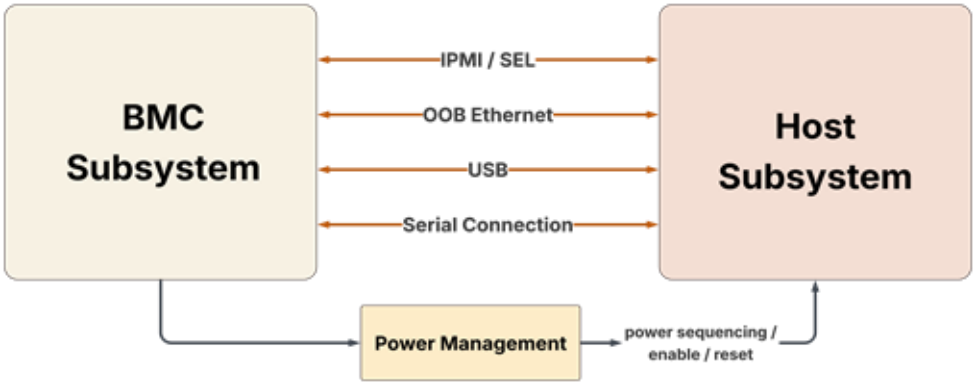
Dual NOS Support

NH-4210

OSPA-Compliant switch platform supporting both FBOSS and SONiC OS.

View NextHop Platforms

Switch Architecture: In-Host Model



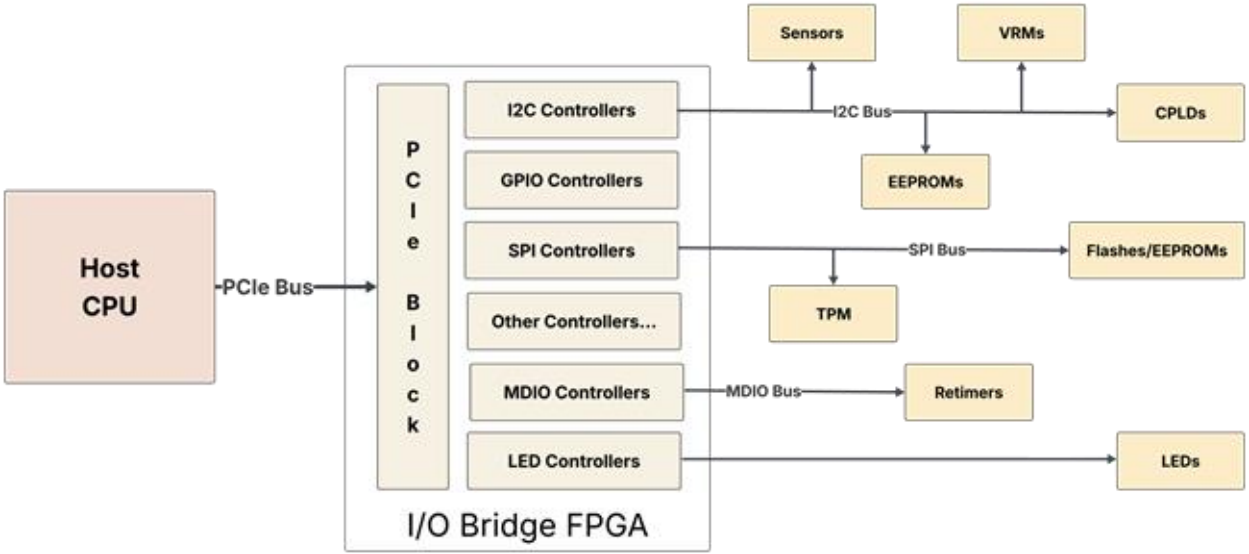
Architectural Decision

OOB-Native vs. In-Host Model

The **In-Host Model** is chosen as the preferred model in the specification.

- Leverages the Host's robust computational resources instead of a constrained BMC.
- Integrates platform management directly into the primary software ecosystem.

Host Subsystem Topology



- Central to this architecture is the I/O Bridge (IOB) FPGA, which serves as the critical aggregation and translation layer between the Host CPU and platform peripherals.

Out-of-band Ethernet Interfaces

01

Dual RJ45 Ports

Reference design supports 2× 1GbE RJ45 front-panel ports.

02

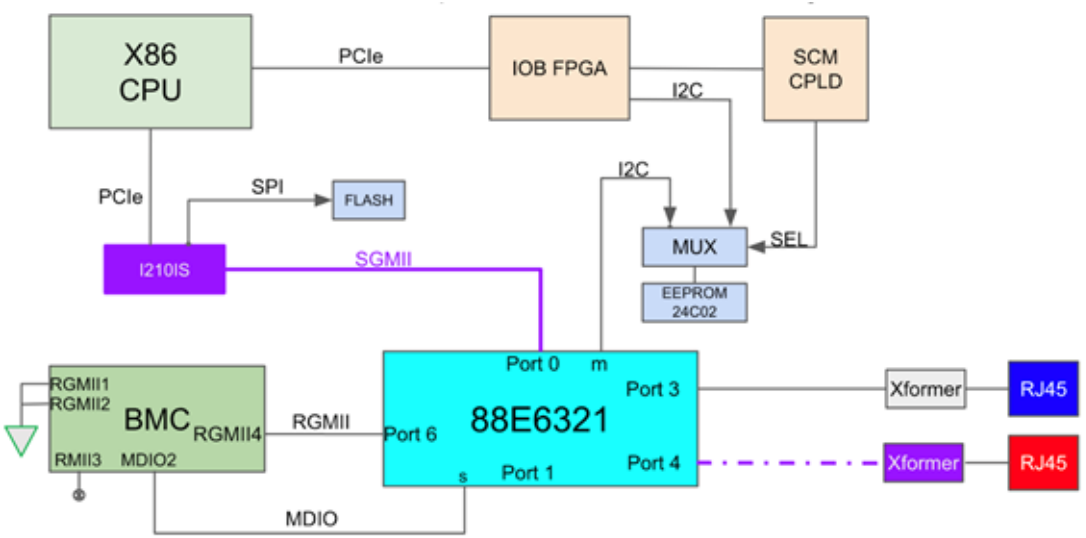
OOB Switch Model

Front-panel ports serve both BMC and Host CPU via an OOB switch.

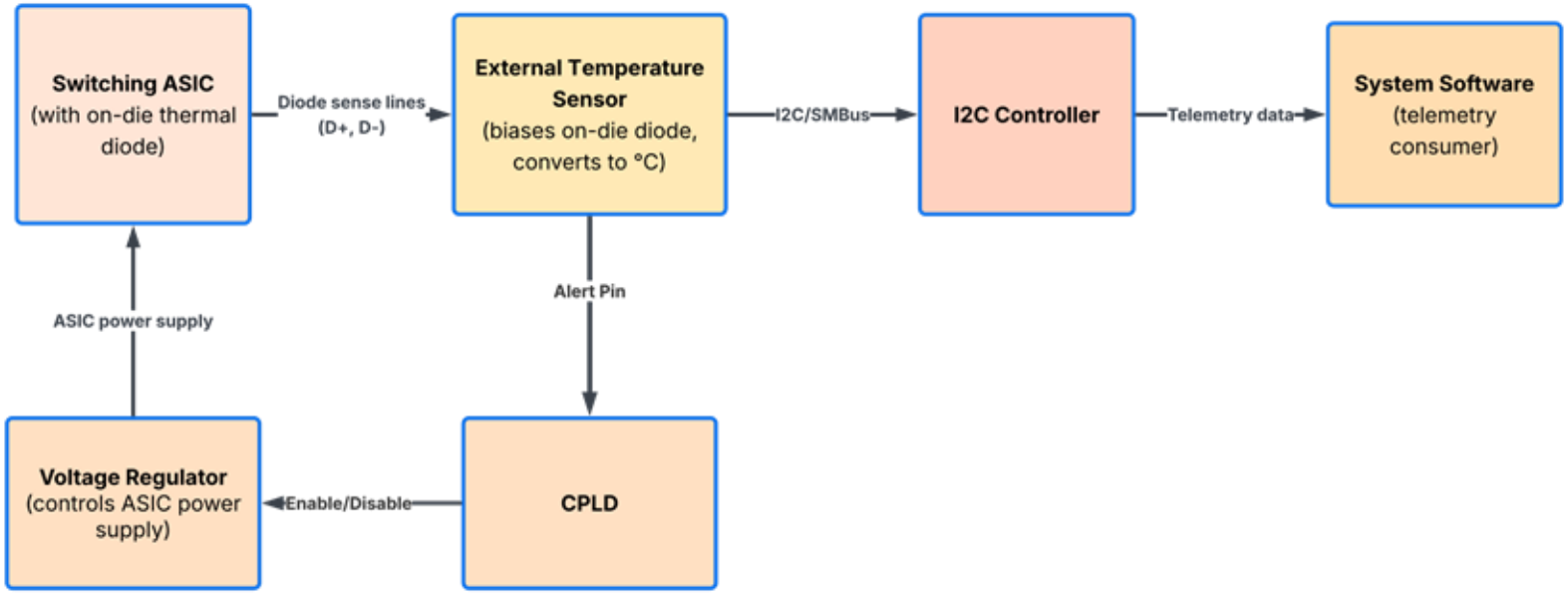
03

EEPROM Programming

OOB Switch EEPROM (containing config) can be programmed in factory and after deployment.



Multi-level ASIC Overheat Protection



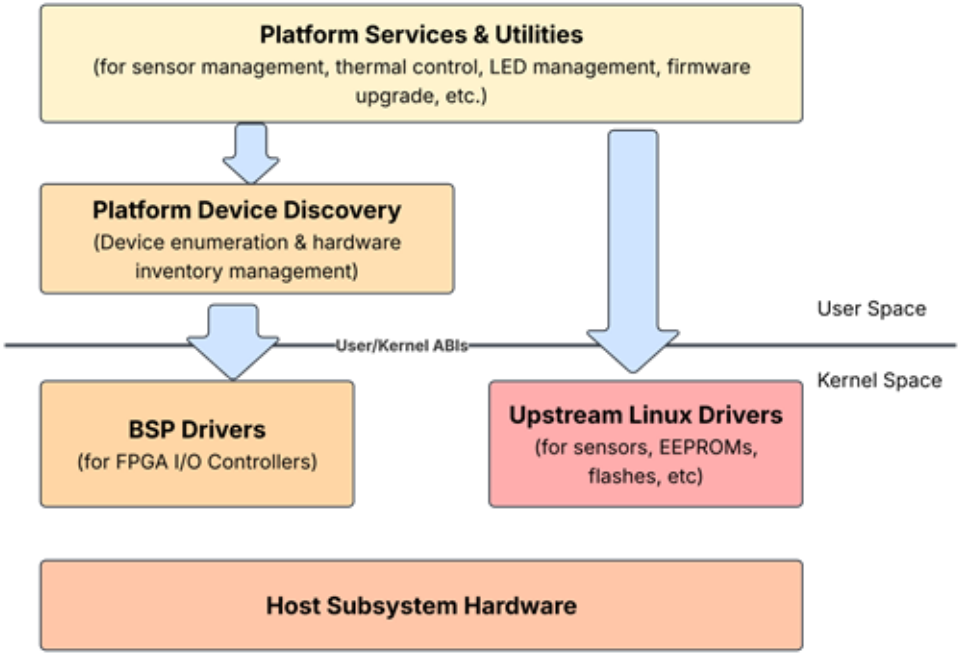
Host (CPU) Software Reference Design

Three-Layer Software Architecture

The standard Switch Platform Architecture organizes the software stack into three distinct layers to maximize modularity and scalability:

- **Device Driver:** Manages unified IOB FPGA controller drivers (for un-upstreamed controllers like I2C/GPIO/SPI) and standard upstream Linux drivers for client devices (sensors, EEPROMs).
- **Platform Device Discovery:** Orchestrates dynamic system discovery and initialization utilizing a configuration-driven approach based on the hierarchical APE topology.
- **Platform Services:** Hosts high-level, hardware-agnostic management services like sensor telemetry, thermal control, LED management, and firmware maintenance.

Key Benefit: Onboarding a new switch platform requires only a new configuration file describing the topology. All services are then automatically initialized.



Call to Action

Join the Initiative

We invite hardware vendors, platform developers, and system architects to collaborate with us in the OCP community:

- https://ocp-all.groups.io/g/ocp-ospa

Improve the Spec

Help refine the OSPA specification by contributing feedback, reviewing proposals, and sharing architectural insights:

- https://github.com/opencomputeproject/OCP-switch-platform-architecture

Adopt & Implement

Integrate OSPA standards into your hardware designs to foster a highly compatible, efficient, and open ecosystem.

Thank You!

